Conditional hybrid retrieval

Execution cost and recall budgets

An indexed cascade study

Research project report · 21 October 2026

The original Blended RAG paper

Keyword, dense and learned-sparse search with hybrid field queries

0.87

Published TREC-COVID nDCG@10

57.63 / 68.40

Published SQuAD answer EM / F1

Published reference values. Our completed study measures retrieval quality and query latency.

What we set out to contribute

1 Measure conditional retrieval under a stated recall budget

2 Separate the BM25 indexing gain from the routing gain

3 Report transfer failures and strong standalone baselines

The project tests an implementation and its limits; it does not introduce a new retrieval principle.

First baseline from our adaptation

Dataset / scope	Queries	r90 retained	Fusion to cascade mean	Time saved
TREC-COVID	50	84.53%	1493.23 to 1440.00 ms	3.56%
SQuAD dev retrieval	10,570	99.38%	148.26 to 139.19 ms	6.12%

In TREC, BM25 took 1410.14 of the 1493.23 ms fusion total.

First Python baseline on overlapping datasets; not the original Elasticsearch / ELSER system.

Reproduction and adaptation boundaries

Dimension	Original paper	This project
Engine / sparse model	Elasticsearch / ELSER	Python / SPLADE
Retrieval question	Blended-query quality	Latency under recall budgets
Answer generation	NQ / SQuAD EM and F1	Not evaluated
Timing reference	Paper system	Our original / indexed fusion

These model, corpus and metric differences limit direct comparison. We did not isolate their individual effects.

Recall retention and absolute recall

Mean method Recall@100
÷ mean fusion Recall@100

TREC primary: 0.11844 ÷ 0.12250 = 96.68%

Hit accuracy: at least one relevant result
Recall: fraction of relevant documents recovered
nDCG: quality of the top-ranked ordering

The live conditional decision

Best Fields BM25, then an 11-feature Ridge gate

Gate condition	Executed action
Predicted gain ≤ threshold	Return BM25 top 100
Predicted gain > threshold	Run Dense + SPLADE, then RRF

Fallback reuses BM25. RRF uses k = 60. Every path returns up to 100 documents.

Four frozen recall budgets

Policy	Threshold	Train OOF retention
r99	0.0629951	99.11%
r95	0.0756657	95.42%
r92	0.0810938	92.59%
r90	0.0847674	90.69%

2,590 NFCorpus train queries. Nominal policy targets are calibration settings, not transfer guarantees.

Completed evaluation scope

Study	Queries	Documents	Timing passes
NFCorpus dev	324	3,633	3
SCIDOCS	1,000	25,657	3
TREC original / indexed	50 / 50	171,332	3 / 3
SQuAD original	10,570	20,958	1 complete
SQuAD indexed	256 fixed	20,958	3

Warmed T4 query timing. The indexed SQuAD run retains the full corpus but samples its questions.

TREC-COVID primary result

1.24×

speedup over indexed fusion

96.68%

Recall@100 retained

91.28 ms indexed fusion / 73.90 ms indexed cascade

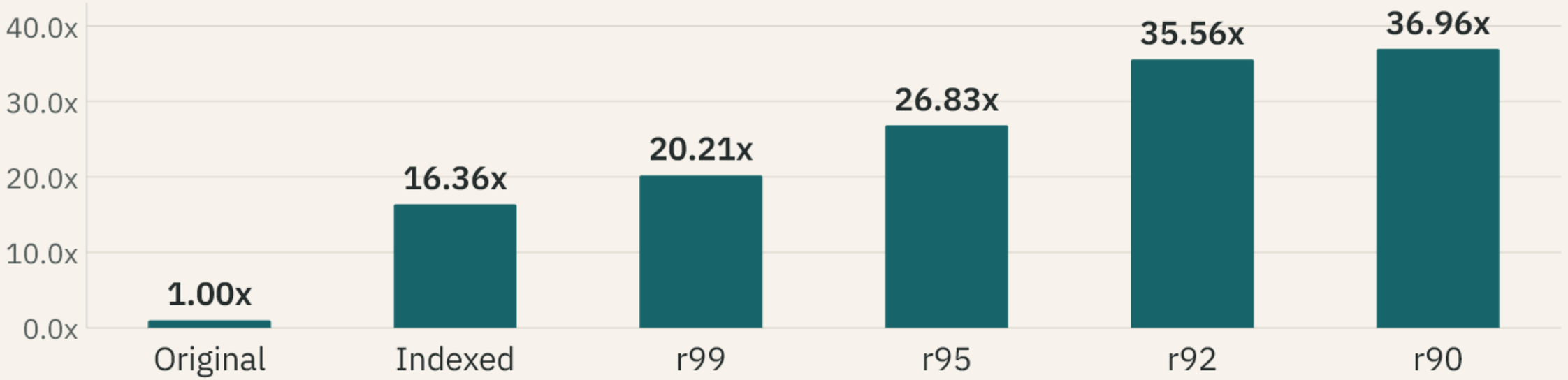
Retention interval: 93.58–98.75%. All 50 queries, full corpus, three warmed passes.

TREC-COVID operating points

Configuration	Retained recall	Mean ms	vs indexed
Indexed fusion	100.00%	91.28	1.00x
r99	96.68%	73.90	1.24x
r95	91.95%	55.66	1.64x
r92	84.53%	41.99	2.17x
r90	84.53%	40.41	2.26x

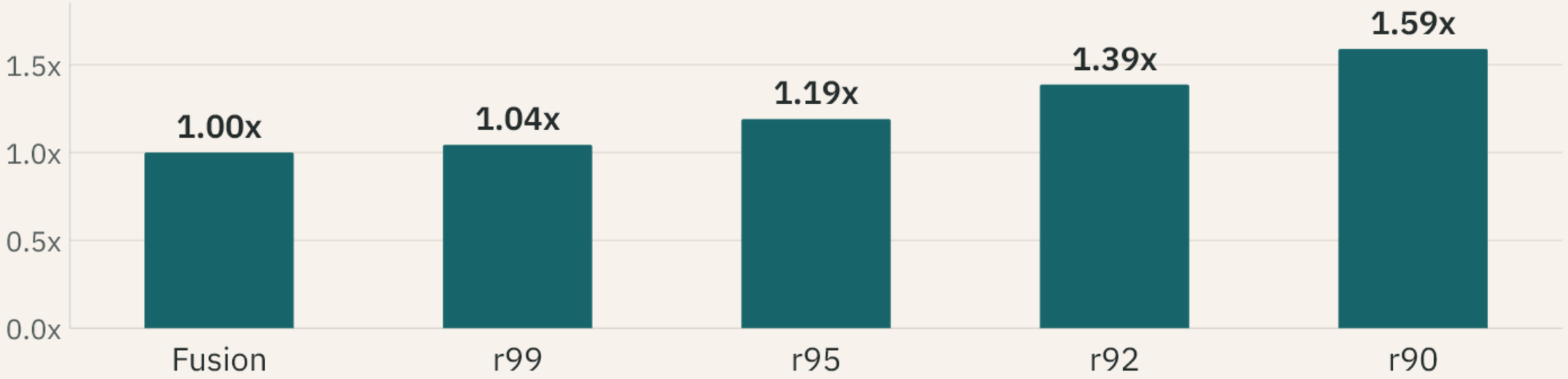
r95: interval crosses 90%. r92 / r90: observed recall falls below 90%. Primary remains r99.

Indexing and routing combined



Retained recall after indexing: 100%, 96.68%, 91.95%, 84.53%, 84.53%. Separate runs include session variation.

SQuAD sampled cascade results



Recall retained: 100%, 100%, 100%, 100%, 99.22%. 256 questions / full corpus / three passes.

Standalone retrievers remain strong

TREC-COVID

Method	Mean ms	Retained
Fusion	91.28	100.00%
BM25	8.11	71.38%
Dense	30.41	69.81%
SPLADE	52.77	104.69%

SQuAD, same sample

Method	Mean ms	Retained
Fusion	29.55	100.00%
BM25	1.03	95.70%
Dense	10.82	95.70%
SPLADE	16.96	98.83%

SPLADE favors TREC recall/time. BM25 already clears the sampled SQuAD 90% floor. Ranking metrics still differ.

Earlier transfer findings

Original backend study	Primary	Retained	Time saved
NFCorpus dev	r92	94.74%	23.00%
SCIDOCS	r92	90.01%	5.53%
TREC-COVID	r90	84.53%	3.56%
SQuAD, full queries	r90	99.38%	6.12%

Original primaries retain their labels. All nominal targets miss on SCIDOCS and TREC-COVID.

Saved evidence and compatibility

400

TREC first-pass rankings

2,048

Sampled SQuAD rankings

Exact IDs, scores, routes and recalls across backends

Independent raw-label scoring reproduces quality. Timing and wider hardware behavior remain experimental limits.

Implementation and reproducibility

File or group	Responsibility
src/indexed_bm25.py	Compatible lexical postings
Frozen routers + indexed protocol	Weights, thresholds, revisions and hashes
Experiment 25	Indexed query execution
Experiments 22 / 29 / 30 / 31	Import, rescore and compare saved evidence
Experiment 32 + publication tools	Rebuild figures and report artifacts

Source paths remain recognizable. Large local inputs are ignored and restored separately when needed.

What to improve in a next study

Full NQ, HotpotQA and CoQA remain deferred

Generated-answer EM/F1 and modern adaptive baselines remain unmeasured

Test new queries under a declared protocol; include hardware, load, energy and index costs.

Overall review

20.21x at 96.68%

Combined gain vs our earlier fusion; includes 1.24x routing over indexed fusion

7.91x at 99.22%

Combined gain vs our earlier fusion; 1.59x vs indexed fusion, 256 questions

A measured execution contribution with explicit quality and baseline limits

Appendix: TREC-COVID quality metrics

Method	Recall@100	nDCG@10	MAP@10	Mean ms	Indexed x	Original x
Fusion	0.1225	0.7632	0.0186	91.28	1.00x	16.36x
BM25	0.0874	0.5497	0.0114	8.11	11.25x	184.02x
Dense	0.0855	0.4725	0.0105	30.41	3.00x	49.10x
SPLADE	0.1282	0.7290	0.0166	52.77	1.73x	28.29x
r99	0.1184	0.7228	0.0177	73.90	1.24x	20.21x
r95	0.1126	0.7056	0.0174	55.66	1.64x	26.83x
r92	0.1036	0.6531	0.0150	41.99	2.17x	35.56x
r90	0.1036	0.6512	0.0150	40.41	2.26x	36.96x

All 50 queries, 171,332 documents, three indexed passes.

Appendix: SQuAD sample quality metrics

Method	Recall@100	nDCG@10	MAP@10	Mean ms	Indexed x	Original x
Fusion	1.0000	0.8093	0.7706	29.55	1.00x	4.98x
BM25	0.9570	0.7787	0.7504	1.03	28.59x	142.25x
Dense	0.9570	0.6878	0.6304	10.82	2.73x	13.58x
SPLADE	0.9883	0.8544	0.8283	16.96	1.74x	8.67x
r99	1.0000	0.8102	0.7708	28.31	1.04x	5.19x
r95	1.0000	0.8028	0.7646	24.82	1.19x	5.92x
r92	1.0000	0.8029	0.7676	21.31	1.39x	6.90x
r90	0.9922	0.8009	0.7676	18.59	1.59x	7.91x

256 fixed questions, 20,958 documents, three indexed passes.

Appendix: primary references

Sawarkar et al. (2024)
Blended RAG

Arabzadeh et al. (2021)
Retrieval strategy selection

Lu (2024)
BM25S eager sparse scoring

Thakur et al. (2021)
BEIR benchmark

Rajpurkar et al. (2016)
SQuAD

Formal et al. (2021) / Reimers & Gurevych (2019)
SPLADE / Sentence-BERT

Full URLs and source details are in speaker notes and the research paper.

Appendix: rebuilding the report

1 Figures from saved reports
experiments/32_prepare_report_assets.py

2 Release artifacts
tools/build_research_materials.py

3 Editable deck and print showcase
tools/build_research_presentation.mjs
tools/render_showcase.mjs

See docs/submission/README.md for exact commands and runtime requirements.